10.8 Green's Functions, the $i\epsilon$ and the Feynman Propagator

LecNote6 evaluates the oscillator path integral exactly,

$$\langle 0|0\rangle_f = \exp\Big[\frac{i}{2}\int dt\,dt'\,f(t)\,G(t-t')\,f(t')\Big],$$
$$G(t) = \int \frac{dE}{2\pi}\,\frac{e^{-iEt}}{-E^2+\omega^2-i\epsilon} = \frac{i}{2\omega}\,e^{-i\omega|t|}, \tag{10.8.1}$$

recognizes G as a Green's function, $(\partial_t^2+\omega^2)G(t)=\delta(t)$ (p. 3), and reads off $\langle 0|TQ(t_1)Q(t_2)|0\rangle = G(t_1-t_2)/i$ (p. 4). That leaves three questions. LecNote4 calls the states made by $a^\dagger$ particles; what are the particles in a quantum-mechanics problem that has only one? Why is G a Green's function, and why this one, which is neither the retarded nor the advanced one? And how are the $(1-i\epsilon)H$ of LecNote5 p. 8, the pole placement on LecNote6 p. 3 and the independence of the result from the initial and final states related? In short: the particles are the quanta of the oscillator. Being a Green's function follows from the equation of motion, in any state; being *this* Green's function is a statement about the vacuum at both ends. The projection onto the vacuum, the pole placement and the Wick rotation are one prescription. The note opens with Green's functions in general, and is the one-mode version of the source discussion in §10.6, Eqs. (10.6.14)–(10.6.16).

Green's functions: from impulse response to propagators

A Green's function solves an inhomogeneous linear differential equation with a point-like source. It is the response of a system to an idealized impulse, and its interpretation changes from classical mechanics to quantum field theory, tied each time to causality.

Classical impulse response. For a driven oscillator, $\ddot{x}(t)+\omega_0^2 x(t) = F(t)/m$, the Green's function $G(t,t')$ is the position at time t produced by a unit impulse at t'. The **retarded** Green's function enforces causality, so that the effect does not precede the cause:

$$G_R(t,t') = \begin{cases} \dfrac{1}{m\omega_0}\sin\big(\omega_0(t-t')\big) & t\geq t', \\ 0 & t<t'. \end{cases}$$

The solution for an arbitrary force is the convolution $x(t)=\int dt'\,G_R(t,t')F(t')$, the sum of the responses to a continuous series of impulses.

The quantum-mechanical kernel. In quantum mechanics the Green's function becomes the **propagator**, or Schrödinger kernel, $K(x,t;x',t') = \langle x|e^{-iH(t-t')}|x'\rangle$ (with $\hbar = 1$). It is the amplitude for a particle to go from (x',t') to (x,t), and, multiplied by $\theta(t-t')$, it solves the Schrödinger equation with a delta-function source: $(i\partial_t - H_x)\,[\theta(t-t')K] = i\,\delta(t-t')\,\delta(x-x')$. It evolves any initial wavefunction,

$$\Psi(x,t) = \int dx'\, K(x,t;x',t')\,\Psi(x',t'),$$

and it is the starting point of the path-integral formulation of quantum mechanics. This kernel is not the same object as the propagator of quantum field theory. The kernel is a Green's function of $i\partial_t - H$ and evolves the wavefunction. The field-theory propagator is a Green's function of the Klein–Gordon operator, built from two-point correlation functions of the field, and it is the subject of the rest of this note.

Causality in quantum field theory. In relativistic QFT, which must account for antiparticles, two Green's functions of the same operator play different roles.

- **The retarded function** G_R, as in the classical case, vanishes for $t < t'$ and describes the response of the field to a source: with a source J, $\langle\phi\rangle = \int G_R\, J$. It is built from the commutator of the field at the two points, which for a free field is a c-number, the same in every state. It embodies the classical notion of cause and effect; it is a response, not a propagation amplitude.
- **The Feynman propagator** G_F describes two processes in one function: a particle propagating forward in time ($t > t'$), and an antiparticle propagating forward in time, which is mathematically the same as a particle propagating *backward* in time ($t < t'$).

The Feynman prescription is a rule for passing the poles in the complex energy plane that yields this. It is what Feynman diagrams use, where an internal line can carry a particle or an antiparticle. The Feynman propagator gives up the purely retarded boundary condition (it is a time-ordered product, not a retarded commutator) in order to describe a quantum theory with particle creation and annihilation. The rest of this note shows, for a single mode, why the vacuum selects it and why the retarded function is a response rather than an amplitude.

The oscillator

Conventions are those of LecNote6: $\hbar = 1$, unit mass, $H = \frac{1}{2}P^2 + \frac{1}{2}\omega^2 Q^2$ with the ground-state energy shifted to $E_0 = 0$ (LecNote5 p. 8), the source enters the Lagrangian as $+fQ$ (LecNote5 p. 7) with f real, and

$\tilde f(E) = \int dt\, e^{iEt} f(t)$. Without the source, in the Heisenberg picture,

$$Q(t) = \frac{1}{\sqrt{2\omega}}\big(a\, e^{-i\omega t} + a^\dagger e^{i\omega t}\big), \qquad [a, a^\dagger] = 1, \qquad H = \omega\, a^\dagger a. \tag{10.8.2}$$

The quanta are the particles

In LecNote5 there is one particle, and Q is its position. Its number is one, trivially and forever. That is not the particle of LecNote4. In LecNote7 the coordinate $Q(t)$ becomes the field $\varphi(\mathbf{x}, t)$, and each mode of the field is an oscillator like this one, with $\omega \to \omega_{\mathbf{k}}$ (LecNote7 p. 1). The particles of LecNote4, created by $a^\dagger(\mathbf{k})$, are the quanta of those oscillators. So Srednicki's §7 is best read as a field theory in $0+1$ dimensions: Q is the field, $a^\dagger$ creates one quantum of energy ω, and $|n\rangle$ is the n-particle state. Read this way, the oscillator is a *free* theory. The potential $\frac{1}{2}\omega^2 Q^2$ is the mass term, not an interaction: the action is quadratic. $N = a^\dagger a$ commutes with H, so the number of particles is conserved, and the only time dependence is the phase $e^{-i\omega t}$ that each quantum carries. That is free propagation, which is all a free particle does. The analogue of the normalization $\langle p|\varphi(0)|0\rangle = 1$ (LecNote4 p. 8) is

$$\langle 1|Q(0)|0\rangle = \frac{1}{\sqrt{2\omega}}, \tag{10.8.3}$$

which is exactly 1 when the one-particle state is normalized as on LecNote4 p. 1, $|k\rangle = \sqrt{2\omega}\,|1\rangle$.

What does change the number of particles is the source. The term $fQ = f\,(a + a^\dagger)/\sqrt{2\omega}$ emits and absorbs quanta: it is the external interaction, and $\langle 0|0\rangle_f$ is the amplitude that the source, switched on and off, leaves no quantum behind. A self-interaction such as Q^3 or Q^4 contains $a^{\dagger 3}$, $a^{\dagger 4}$, …, and with it N is no longer conserved.

Eq. (10.8.1) gives the probability directly. Since $|e^{iX/2}|^2 = e^{-\operatorname{Im} X}$ and $\operatorname{Im} G(t) = \cos(\omega t)/2\omega$,

$$\big|\langle 0|0\rangle_f\big|^2 = \exp\Big[-\frac{|\tilde f(\omega)|^2}{2\omega}\Big]. \tag{10.8.4}$$

The source leaves the oscillator in a coherent state, with a Poisson-distributed number of quanta of mean $\bar n = |\tilde f(\omega)|^2/2\omega$, and (10.8.4) is $P(0) = e^{-\bar n}$. Only the component of f at frequency ω enters: a quantum is emitted only if the source can supply its energy. Eq. (10.6.15) is the same statement for a field. As a check, take $f = 0.8\, e^{-t^2/2s^2}$ with $s = 0.8$ and $\omega = 1$, so $\bar n = 0.6785$. Evolving $|0\rangle$ with $H - fQ$ in a truncated Fock space gives $\langle N\rangle = 0.6785$ afterwards, occupation probabilities 0.5074, 0.3443, 0.1168, 0.0264 for $n = 0, \dots, 3$ (the Poisson values), and $\langle 0|U|0\rangle = 0.65981 + 0.26837\,i$, which is (10.8.1) to all the digits shown.

Every time-ordered two-point function is a Green's function

The Green's-function property on LecNote6 p. 3 needs neither the path integral nor the vacuum. Take any normalized state $|\psi\rangle$. Differentiating the θ-functions of the time ordering gives delta functions times equal-time commutators. With $\dot{Q} = P$, $\dot{P} = -\omega^2 Q$ and $[Q(t), P(t)] = i$,

$$\begin{aligned} \partial_t \langle\psi|TQ(t)Q(t')|\psi\rangle &= \langle\psi|TP(t)Q(t')|\psi\rangle + \delta(t-t')\,\langle\psi|[Q(t),Q(t)]|\psi\rangle, \\ (\partial_t^2+\omega^2)\,\langle\psi|TQ(t)Q(t')|\psi\rangle &= \delta(t-t')\,\langle\psi|[P(t),Q(t)]|\psi\rangle = -i\,\delta(t-t'). \end{aligned} \tag{10.8.5}$$

So $i\langle\psi|TQ(t)Q(t')|\psi\rangle$ is a Green's function of the oscillator, whatever $|\psi\rangle$ is. With a potential $V(Q)$ added, the right-hand side gains $-\langle\psi|T\,V'(Q(t))\,Q(t')|\psi\rangle$, but the contact term $-i\delta(t-t')$ is always there. This is the simplest Schwinger–Dyson equation; Eq. (10.6.21) is its field-theory version.

For a number eigenstate, (10.8.2) gives

$$\begin{aligned} \langle n|TQ(t)Q(0)|n\rangle &= \frac{G(t)}{i} + \frac{n\cos\omega t}{\omega}, \\ \int dt\, e^{iEt}\,\langle n|TQ(t)Q(0)|n\rangle &= \frac{i}{E^2-\omega^2+i\epsilon} + 2\pi n\,\delta(E^2-\omega^2). \end{aligned} \tag{10.8.6}$$

The extra term solves the homogeneous equation, so (10.8.5) still holds, but this is a different Green's function. In energy it is an on-shell delta function. In a thermal state n is replaced by the Bose factor, and (10.8.6) becomes the time-ordered component of the real-time propagator at finite temperature.

The same goes for the commutator. For this H, $[Q(t),Q(0)] = -i\sin(\omega t)/\omega$ is a c-number, the same in every state, and both

$$\begin{aligned} G_R(t) &= i\theta(t)\,\langle\psi|[Q(t),Q(0)]|\psi\rangle = \theta(t)\,\frac{\sin\omega t}{\omega}, \\ G_A(t) &= -i\theta(-t)\,\langle\psi|[Q(t),Q(0)]|\psi\rangle = -\theta(-t)\,\frac{\sin\omega t}{\omega} \end{aligned} \tag{10.8.7}$$

satisfy $(\partial_t^2+\omega^2)G = \delta$.

The equation of motion fixes a Green's function $G(t-t')$ only up to $Ae^{i\omega t} + Be^{-i\omega t}$. Being a Green's function is the dynamics. *Which* Green's function is a boundary condition. G_R and G_A are the same in every state; for the time-ordered function the boundary condition is the state.

Which one: the Feynman Green's function

In energy the three Green's functions differ only in which way the poles at $E = \pm\omega$ are pushed off the real axis (table on p. 55, Fig. 10.2). The

retarded function closes below for $t > 0$ and picks up both poles; for $t < 0$ it closes above and finds none. The Feynman function picks up $+\omega$ for $t > 0$ and $-\omega$ for $t < 0$.

	poles	$G(t)$	operator form
retarded	$\pm\omega - i\epsilon$	$\theta(t)\,\sin(\omega t)/\omega$	$i\theta(t)\langle\psi\|[Q(t),Q(0)]\|\psi\rangle$, any ψ
advanced	$\pm\omega + i\epsilon$	$-\theta(-t)\,\sin(\omega t)/\omega$	$-i\theta(-t)\langle\psi\|[Q(t),Q(0)]\|\psi\rangle$, any ψ
Feynman	$\omega - i\epsilon,\ -\omega + i\epsilon$	$(i/2\omega)\,e^{-i\omega\|t\|}$	$i\langle 0\|TQ(t)Q(0)\|0\rangle$, vacuum only

THE THREE GREEN'S FUNCTIONS OF $\partial_t^2 + \omega^2$

The three Green's functions met in these notes. All satisfy $(\partial_t^2 + \omega^2)G = \delta(t)$.

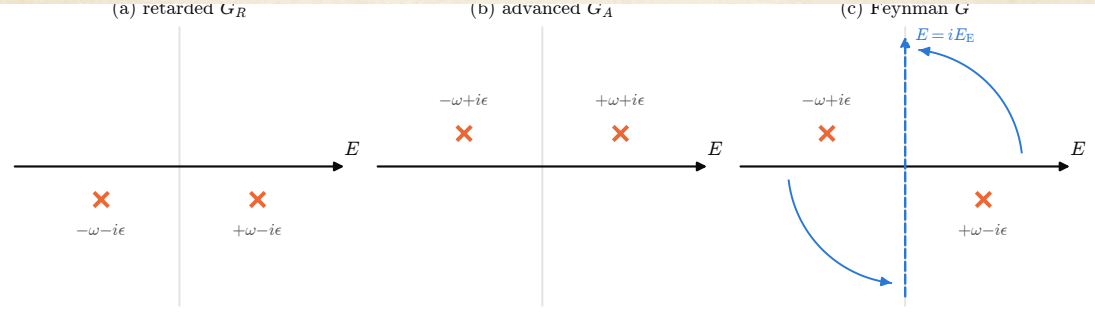


Figure 10.2: Pole placements in the complex E plane. (c) also shows the Wick rotation $E = iE_{\rm E}$, which sweeps the first and third quadrants. It crosses no pole only for the Feynman placement.

Reading it off the operators. From (10.8.2), $\langle 0|Q(t)Q(t')|0\rangle = e^{-i\omega(t-t')}/2\omega$. The right-hand Q creates a quantum at t', the left-hand Q absorbs it at t, and in between the state carries energy ω. Time ordering puts the later time on the left, so whichever of t, t' is earlier, the quantum is created then:

$$G(t) = \frac{i}{2\omega}\big[\theta(t)\,e^{-i\omega t} + \theta(-t)\,e^{+i\omega t}\big]. \tag{10.8.8}$$

Positive frequency propagates forward in time and negative frequency backward. Equivalently, positive energy always flows from the earlier time to the later one. The vacuum is the bottom of the spectrum, so an operator acting on it can only add energy; it can never borrow any. That is the content of the Feynman prescription. For a complex field the backward-going negative-frequency part is read as an antiparticle going forward, as in the overview above. For the real Q here it is the same quantum, created at the earlier of the two times.

Three decompositions. From (10.8.1) and (10.8.7),

$$G(t) = G_R(t) + \frac{i}{2\omega}\,e^{i\omega t} = G_A(t) + \frac{i}{2\omega}\,e^{-i\omega t}, \tag{10.8.9}$$

$$\mathrm{Re}\,G(t) = \tfrac{1}{2}\big[G_R(t) + G_A(t)\big] = \frac{\sin\omega|t|}{2\omega}, \qquad \mathrm{Im}\,G(t) = \tfrac{1}{2}\langle 0|\{Q(t),Q(0)\}|0\rangle = \frac{\cos\omega t}{2\omega}. \tag{10.8.10}$$

The Feynman function is the retarded one plus a solution of the free equation. Its real part is built from the commutator alone, so it is the same in every state: it is the dynamics. Its imaginary part is the symmetrized vacuum correlation; at $t = 0$ it is $\langle 0|Q^2|0\rangle = 1/2\omega$, the zero-point spread. In $|n\rangle$ it becomes $(2n+1)\cos(\omega t)/2\omega$, which is (10.8.6) again. So the Feynman function is the dynamics plus the vacuum's fluctuations, and the state sits entirely in the homogeneous part. In energy, $1/(-E^2+\omega^2-i\epsilon) = \mathcal{P}\,1/(\omega^2-E^2) + i\pi\,\delta(E^2-\omega^2)$: the imaginary part is on shell, and it is what makes $|\langle 0|0\rangle_f| < 1$ in (10.8.4).

An amplitude, not a response

Why is the answer not the retarded function, the one classical mechanics uses? Because $\langle 0|0\rangle_f$ answers a different question.

Response. Start in $|0\rangle$ and push the oscillator with f. With $H - fQ$ the Heisenberg equation $\ddot{Q} + \omega^2 Q = f$ holds as an operator statement, so the expectation value obeys the classical equation, with $\langle Q\rangle = \langle P\rangle = 0$ in the past:

$$\langle Q(t)\rangle = \int dt'\, G_R(t-t')\, f(t'). \tag{10.8.11}$$

It is real and causal. It is an in–in quantity: the same state on both sides.

Amplitude. $\langle 0|0\rangle_f = \langle 0_{\rm out}|0_{\rm in}\rangle_f$ is the amplitude to start in the vacuum *and end* in the vacuum (§10.6). Its first derivative is an in–out matrix element,

$$q_{\rm io}(t) \equiv \frac{1}{i}\frac{\delta \ln\langle 0|0\rangle_f}{\delta f(t)} = \frac{\langle 0_{\rm out}|Q(t)|0_{\rm in}\rangle_f}{\langle 0_{\rm out}|0_{\rm in}\rangle_f} = \int dt'\, G(t-t')\, f(t'), \tag{10.8.12}$$

the one-mode version of Eq. (10.6.16). It is complex, and it is not zero before the source is switched on. That is not acausal: $q_{\rm io}$ is conditioned on ending in the vacuum, and it is not an observable. The vacuum is imposed at both ends, so this is a boundary-value problem with conditions at $t \to -\infty$ *and* $t \to +\infty$, not an initial-value problem. By (10.8.8), $q_{\rm io}$ contains only $e^{-i\omega t}$ after the source has acted and only $e^{+i\omega t}$ before. Those are the two conditions. With the source on, define $a(t) = e^{i\omega t}\big[\sqrt{\omega/2}\,Q(t) + iP(t)/\sqrt{2\omega}\big]$, which is $a_{\rm in}$ before the source and $a_{\rm out}$ after it. Then $a_{\rm in}|0_{\rm in}\rangle = 0$ removes the a part early, and $\langle 0_{\rm out}|a^\dagger_{\rm out} = 0$ removes the $a^\dagger$ part late.

Fig. 10.3 and the table on p. 57 show both for a Gaussian pulse. The operator columns evolve $|0\rangle$ forward and $\langle 0|$ backward in a truncated Fock space, which does by hand what the $i\epsilon$ does in the path integral. After the pulse, $\langle Q\rangle = 2\,\mathrm{Re}\,q_{\rm io}$ (to 5×10^{-7}): the in–out quantity keeps only the positive-frequency half of the response. Before the pulse $\langle Q\rangle = 0$, but $q_{\rm io} = (i/2\omega)\,e^{i\omega t}\tilde{f}(\omega)^* \neq 0$.

	in–in $\langle Q(t)\rangle$		in–out $q_{\mathrm{io}}(t)$	
ωt	operators	$\int G_R f$	operators	$\int G f$
-3.0	0.00000	0.00000	$0.00514 - 0.03604\,i$	$0.00514 - 0.03604\,i$
0.0	0.02597	0.02597	$0.02597 + 0.03640\,i$	$0.02597 + 0.03640\,i$
1.5	0.07351	0.07351	$0.03719 + 0.00258\,i$	$0.03719 + 0.00258\,i$
4.0	-0.05510	-0.05510	$-0.02755 - 0.02380\,i$	$-0.02755 - 0.02380\,i$
7.0	0.04783	0.04783	$0.02392 + 0.02744\,i$	$0.02392 + 0.02744\,i$

RESPONSE AND IN–OUT VALUE FOR A GAUSSIAN PULSE

$f = 0.05\,e^{-t^2/2s^2}$, $s = 0.8$, $\omega = 1$. "Operators": time evolution with $H - fQ$ in an 80-state Fock space. The convolutions use (10.8.7) and (10.8.1).

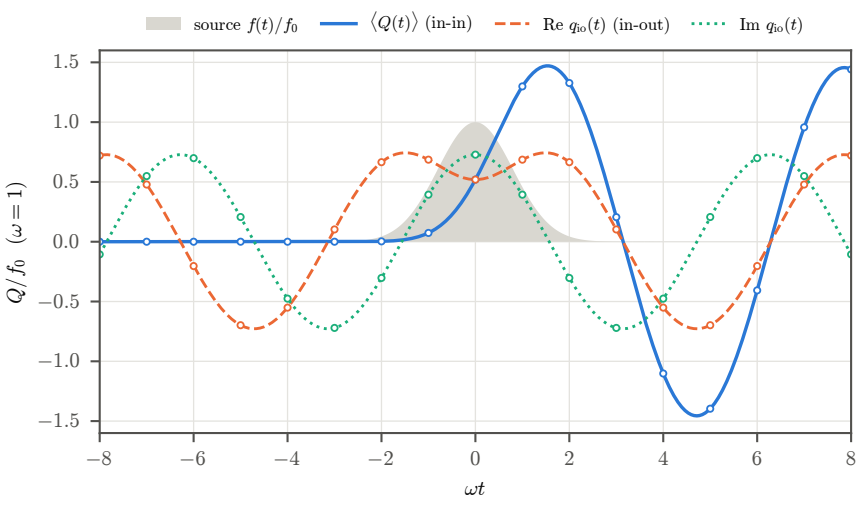


Figure 10.3: The pulse of the table on p. 57. Lines: operator calculation. Circles: $\int G_R f$ and $\int G f$. The response $\langle Q\rangle$ is zero until the pulse arrives; the in–out value is not.

So G should not be read as the way the position responds to a kick. At $f = 0$ both G and G_R are vacuum correlators, a time-ordered product and a retarded commutator. With a source on, G is the kernel of the in–out amplitude and G_R is the kernel of the response. LSZ needs the in–out kernel: LecNote4 builds the S-matrix from vacuum-to-vacuum time-ordered products. Expectation values in a driven state call for an in–in (closed-time-path, or Schwinger–Keldysh) generating functional, which this course does not use.

The $(1 - i\epsilon)$ prescription: one $i\epsilon$, three faces

The factor $1 - i\epsilon$ is put into the Hamiltonian (LecNote5 p. 8) for two reasons: it makes the oscillatory path integral converge, by damping high-energy configurations with $e^{-\epsilon HT}$, and it projects the initial and final states onto the vacuum, so that the path integral computes the vacuum-to-vacuum amplitude whatever states were actually prepared. Replacing $H \to (1 - i\epsilon)H$ in e^{-iHT} is the same as $T \to T(1 - i\epsilon)$: time is tilted slightly into the lower half plane. That one change does three things, which are the same thing.

It projects onto the vacuum. With $c_n = \langle n|\psi\rangle$ and elapsed time $T \to +\infty$, on the tilted line

$$e^{-iHT(1-i\epsilon)}|\psi\rangle = \sum_n c_n\, e^{-iE_nT} e^{-\epsilon E_n T}|n\rangle \;\to\; c_0|0\rangle \tag{10.8.13}$$

as $T \to \infty$ at fixed ϵ, i.e. for $\epsilon T \gg 1$, with $\epsilon \to 0$ taken last (LecNote5 p. 8). The same happens at the far end. The two overlaps with $|0\rangle$, which must both be nonzero, are removed by $\langle 0|0\rangle_{f=0} = 1$. This needs a unique ground state; with a degenerate one the projection is onto the ground subspace, and the result does depend on the preparation. A gap makes the approach exponential, $e^{-\epsilon\omega T}$ for the oscillator.

It places the poles. In the path integral the same tilt multiplies $\frac{1}{2}\dot{q}^2$ by $1 + i\epsilon$ and $\frac{1}{2}\omega^2 q^2$ by $1 - i\epsilon$ (LecNote6 p. 1), so the kernel is $(1 + i\epsilon)E^2 - (1 - i\epsilon)\omega^2 = E^2 - \omega^2 + i\epsilon(E^2 + \omega^2)$. The coefficient of ϵ is positive, so this is $E^2 - \omega^2 + i\epsilon$ (p. 2), with poles at $E = \pm(\omega - i\epsilon')$, $\epsilon' = \epsilon/2\omega$ after the rescaling: the Feynman placement. The same can be seen in time. On the tilted line $|G(t(1 - i\epsilon))| = e^{-\epsilon\omega|t|}/2\omega$ decays in both directions, while G_R grows like $e^{\epsilon\omega t}$ for $t > 0$ and G_A like $e^{\epsilon\omega|t|}$ for $t < 0$. "The vacuum at both ends" and "the Feynman poles" are the same requirement: the Green's function that stays bounded when time is tilted.

It is the start of the Wick rotation. Tilt all the way, $t = -i\tau$. Then $\partial_t^2 + \omega^2 \to -\partial_\tau^2 + \omega^2$, a positive operator, and its Green's function has no freedom left:

$$(-\partial_\tau^2 + \omega^2)\, G_{\rm E}(\tau) = \delta(\tau), \qquad G_{\rm E}(\tau) = \frac{e^{-\omega|\tau|}}{2\omega}, \qquad G(-i\tau) = i\, G_{\rm E}(\tau). \tag{10.8.14}$$

Adding $Ae^{\omega\tau} + Be^{-\omega\tau}$ would make G_{E} blow up at one end, so the bounded Euclidean Green's function is unique, and G is its analytic continuation. In energy, the contour can be rotated to $E = iE_{\mathrm{E}}$ without crossing a pole only when the poles sit in the second and fourth quadrants (Fig. 10.2c; the pole sketch on LecNote6 p. 3 has them there). With the retarded or advanced placement one pole is always in the way.

It is tempting to read the logic backwards: the result is a Green's function, so it cannot depend on how the states were prepared, and that justifies the $i\epsilon$. But every state gives a Green's function, (10.8.5). Being a Green's function is not what removes the preparation; the projection does. The projection is the $i\epsilon$, and the same $i\epsilon$, seen in energy, is the contour. Nor is independence of the preparation independence of the state: the $i\epsilon$ replaces whatever was prepared by the vacuum, for any preparation that overlaps it. With $|n\rangle$ at both ends the answer would be the different Green's function (10.8.6).

Adiabatic switching. In scattering theory, interactions are often said to be switched off adiabatically at early and late times to define in and out states. The $i\epsilon$ is related to this but is not the same thing. It does not switch the interaction off: it multiplies all of H, free and interacting parts alike, and projects onto the ground state of the full Hamiltonian, the interacting vacuum. Adiabatic switching reaches the same state from the other side: by the Gell-Mann–Low theorem, switching the interaction on slowly carries the free vacuum into an eigenstate of the full Hamiltonian, when that limit exists. In the operator route of Srednicki's Problem 9.5(j) the factor $(1 - i\epsilon)$ multiplies H_0 alone, not all of H as in (6.21); there it projects onto the free vacuum $|\emptyset\rangle$, and the denominator of (9.41) fixes the normalization $|\langle\emptyset|0\rangle|^2$.

What it does for scattering.

- **Vacuum selection.** Only the ground state survives at both ends, so the result does not depend on the states prepared, provided they overlap with the vacuum.
- **Vacuum-to-vacuum correlators.** With the vacuum at both ends, the path integral with sources computes $Z[J] = \langle 0|0\rangle_J$ and, by differentiation, vacuum expectation values of time-ordered products, with the Feynman propagator as the free two-point function.
- **LSZ.** The LSZ formula expresses S-matrix elements through exactly these time-ordered vacuum correlators (§10.6). The particles of the in and out states enter through LSZ; the $i\epsilon$ supplies the vacuum they are built on.

From one oscillator to the free field

In LecNote7 each mode of φ is an oscillator with $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$, and the same steps give the Feynman propagator $\Delta(x-x') = \int \frac{d^4k}{(2\pi)^4}\, e^{ik(x-x')}/(k^2+m^2-i\epsilon)$, whose k^0 integral is (10.8.1) with $\omega \to \omega_{\mathbf{k}}$ (LecNote7 p. 3). Everything above holds mode by mode. The quanta of mode k are particles of momentum k. $|\langle 0|0\rangle_J|^2$ is the probability that the source emits nothing, and $-\ln|\langle 0|0\rangle_J|^2$ is the mean number it emits (Eq. (10.6.15)). $\mathrm{Re}\,\Delta$ comes from the commutator and $\mathrm{Im}\,\Delta$ from the vacuum's fluctuations. The in–in response to J is the retarded function, and the in–out value is $\int \Delta J$.

Off shell is the time ordering. The d^4k form of Δ integrates over every k^μ, on shell or not, but the off-shell k are not extra modes. Doing the k^0 integral leaves only on-shell ones, (10.8.8) mode by mode:

$$\Delta(x-x') = i\int \widetilde{dk}\,\big[\theta(t-t')\,e^{ik(x-x')} + \theta(t'-t)\,e^{-ik(x-x')}\big], \qquad k^0 = \omega_{\mathbf{k}}, \tag{10.8.15}$$

with $\widetilde{dk} = d^3k/\big((2\pi)^3 2\omega_{\mathbf{k}}\big)$. A function of t with only the frequencies $\pm\omega$ is $Ae^{-i\omega t} + Be^{i\omega t}$, which solves the free equation and is not a Green's function. The $\delta(t)$ requires a jump in $\dot G$, so $G = (i/2\omega)\,e^{-i\omega|t|}$ has a kink at $t=0$, and the kink spreads it over every energy; that spread is $1/(-E^2+\omega^2-i\epsilon)$. So the off-shell k^0 in Δ are the Fourier content of the two θ-functions in (10.8.15), that is, of the time ordering. The quanta are on shell; the off-shell k^0 record only that each is created at the earlier of the two times and absorbed at the later. In the path integral the off-shell components are integration variables: every Fourier component $\tilde\varphi(k)$ of every field configuration enters with its phase, and $\tilde\Delta(k)/i$ is its two-point function. A configuration with $k^2+m^2 \neq 0$ does not solve the free field equation. It is a fluctuation, not a wave that can travel.

Outside the light cone. The split (10.8.10) carries over mode by mode. With the c-numbers $\Delta_R(x-x') = i\theta(t-t')\,[\varphi(x),\varphi(x')]$ and $\Delta_A(x-x') = -i\theta(t'-t)\,[\varphi(x),\varphi(x')]$,

$$\Delta(x-x') = \tfrac12\big[\Delta_R+\Delta_A\big](x-x') + \tfrac{i}{2}\,\langle 0|\{\varphi(x),\varphi(x')\}|0\rangle. \tag{10.8.16}$$

The first term is real and the same in every state; the second is i times the symmetrized vacuum correlation. For spacelike $x-x'$ the commutator vanishes (microcausality), so $\Delta_R = \Delta_A = 0$ there and

$$\Delta(x-x') = i\,\langle 0|\varphi(x)\varphi(x')|0\rangle = \frac{i\,m}{4\pi^2 r}\,K_1(mr), \qquad r=\sqrt{(x-x')^2}>0,$$
$$\Delta(x-x') \;\to\; \frac{i}{2(2\pi)^{3/2}}\sqrt{\frac{m}{r^3}}\,e^{-mr} \qquad (mr \gg 1). \tag{10.8.17}$$

Outside the light cone Δ is purely the correlation of the vacuum's zero-point fluctuations at two points, the field version of $\langle 0|Q^2|0\rangle = 1/2\omega$, with correlation length $1/m$. It is a correlation, not a propagation. A chain of masses joined by springs of stiffness K, each also tied to its rest position by a spring k_0, behaves the same way: in the ground state the displacements of distant masses are correlated, and the correlation falls off over $\sqrt{K/k_0}$ lattice spacings when $k_0 \ll K$, the lattice version of $1/m$. No signal is involved. A source switched on near x' changes $\langle\varphi(x)\rangle$ by $\int \Delta_R J$, the in–in response, which vanishes outside the light cone of the source. The in–out value $\int \Delta J$ does not, for the reason $q_{\rm io}$ is nonzero before the pulse: it is conditioned on ending in the vacuum, and it is not an observable.

Off shell, with a source. Off-shell momenta do carry physics when a source is on, as a driven field rather than a free one. By (10.8.4) and its field version (10.6.15), quanta are emitted only by the on-shell part of $\tilde{J}$; the off-shell part drives a field that stays with the source. A static source, $J = g\,\delta^3(\mathbf{x})$, is the extreme case. Its transform sits at $k^0 = 0$, where $k^2 + m^2 = \mathbf{k}^2 + m^2$ never vanishes, so it emits nothing, the $i\epsilon$ plays no role, and $\int \Delta J$ and $\int \Delta_R J$ agree:

$$\langle\varphi(\mathbf{x})\rangle = g \int \frac{d^3k}{(2\pi)^3}\,\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{\mathbf{k}^2 + m^2} = \frac{g\,e^{-mr}}{4\pi r}, \qquad r = |\mathbf{x}|. \tag{10.8.18}$$

This is the Yukawa field, and it goes nowhere. An internal line of a diagram is off shell in the same sense: its momentum is fixed at the vertices, and it carries the field one vertex drives at another, not a quantum that escapes.

LecNote7 p. 1 writes the prescription as $m^2 \to (1 - i\epsilon)m^2$. In fact $(1 - i\epsilon)H_0$ damps all three terms of H_0 and gives the kernel $-(k^2 + m^2) + i\epsilon(k_0^2 + \mathbf{k}^2 + m^2)$, whose ϵ coefficient is positive. So the general form is $\omega_{\mathbf{k}}^2 \to \omega_{\mathbf{k}}^2 - i\epsilon$ in every mode, i.e. $m^2 \to m^2 - i\epsilon$. Writing $(1 - i\epsilon)m^2$ is equivalent for $m \neq 0$ but gives nothing at $m = 0$, where $k^2 - i\epsilon$ is still the right prescription.

At $m = 0$ there is no gap. In quantum mechanics the limit $\omega \to 0$ is a free particle, which has no normalizable ground state, and $G \propto 1/\omega$ diverges. In the massless field $\omega_{\mathbf{k}} \to 0$ only as $\mathbf{k} \to 0$, and in $3 + 1$ dimensions the soft modes carry little weight: $\int d^3k/2\omega_{\mathbf{k}} \sim \int k\,dk$ converges, so Δ has no infrared divergence. (In $1 + 1$ dimensions it does.) The projection onto the Fock vacuum still works, but its approach is a power of T rather than exponential.

Summary

- The particles of the oscillator are its quanta. The oscillator is the free theory of one mode, and the source f is the external

interaction that emits and absorbs them.

- Every time-ordered two-point function is a Green's function, in any state. That is the equation of motion. Which Green's function it is depends on the state.
- The vacuum at both ends picks the Feynman function: positive energy flows forward in time, $G = G_R + (i/2\omega)e^{i\omega t}$, $\mathrm{Re}\, G$ is the dynamics and $\mathrm{Im}\, G$ the vacuum's fluctuations.
- With a source on, G is the kernel of the in–out amplitude, and G_R is the kernel of the response of $\langle Q \rangle$. LSZ needs the former.
- $(1 - i\epsilon)H$ tilts time. The projection onto the vacuum, the Feynman poles and the Wick rotation are one prescription.
- The modes in Δ are on shell; its off-shell k^0 are the Fourier content of the time ordering. Outside the light cone $\Delta_R = \Delta_A = 0$ and $\Delta = i\langle 0|\varphi(x)\varphi(x')|0\rangle \sim e^{-mr}$: a vacuum correlation, not a signal. Off-shell momenta are physical as the driven field of a source, such as the Yukawa field of a static one.